Data Science

IA and Ethics

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Outline

1. Introduction
2. Privacy
3. Fairness

3 different perspectives

Privacy: Data are owned by users ; they might be used collectively (estimation, inference) but cannot be traced back to specific user

Fairness: Users are different but should be treated equally; some features are sensible and should not impact the outcome of an algorithm

Security: Users are willing to send and share data; yet outsiders should not be able to decypher them. Related to encryption (homomorphic ?)

Reference: This lecture is based on the lecture from Vianney Perchet (ENSAE) and his website

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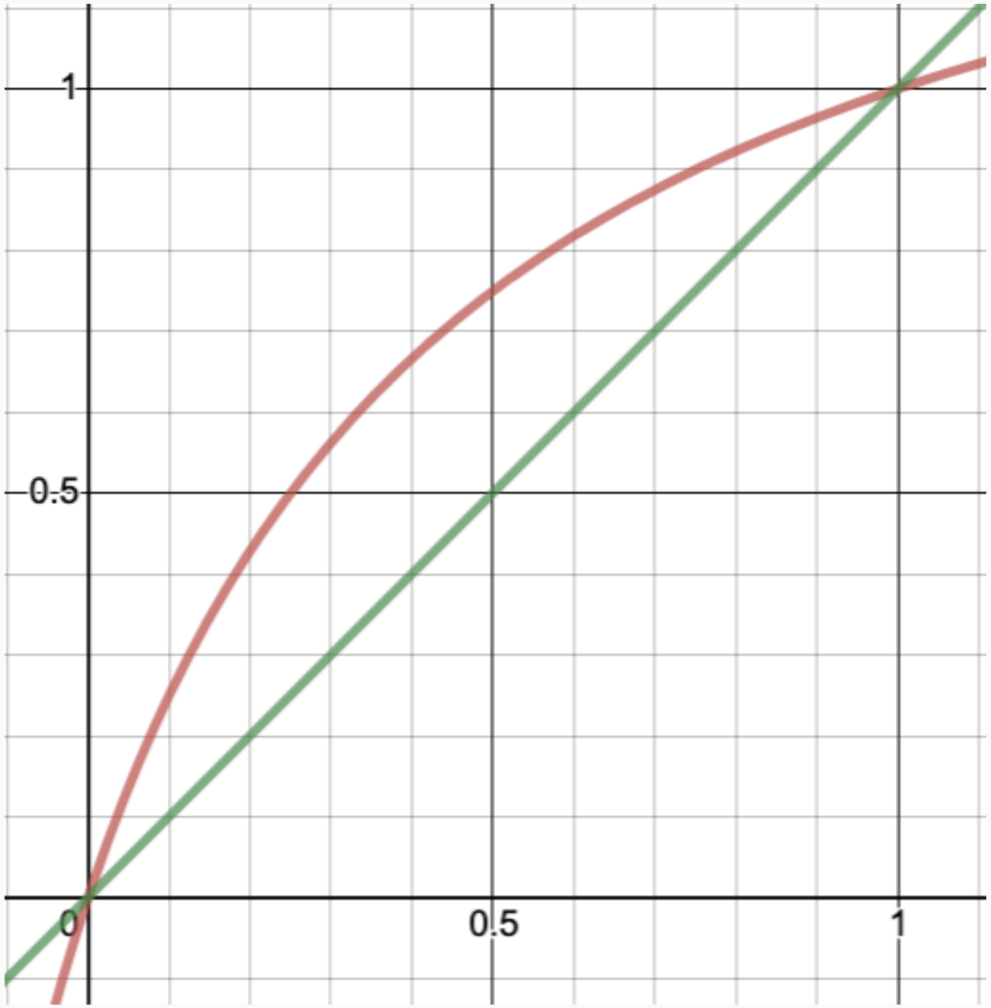
A simple example (Local Differential Privacy)

- Assume $X_i \in \{0, 1\}$, Bernoulli parameter $p \approx \frac{1}{n}\sum_{i=1}^{n} X_i$
- **Construct** dataset to learn p while keeping each X_i "**private**"
- $\tilde{X}_i = X_i$ with proba $1 - \eta = 0.5$ and random w.p. $\eta = 0.5$
- $\mathbb{E}\left[\tilde{X}_i\right] = \frac{1}{4} + \frac{p}{2}$ so that $\hat{p} = 2\frac{1}{n}\sum_{i=1}^{n} \tilde{X}_i - \frac{1}{2}$

$$\mathbb{P}(X_i = 1 \mid \tilde{X}_i = 1) = \frac{3p}{1+2p} = p + (1-p)(1 - \frac{1}{1+2p})$$

  - If $p = 1$ then $\mathbb{P}(X_i = 1 \mid \tilde{X}_i = 1) = 1$
  - If $p = \frac{1}{2}$ then $\mathbb{P}(X_i = 1 \mid \tilde{X}_i = 1) = \frac{3}{4}$
  - If $p = 1 - \epsilon$ then $\mathbb{P}(X_i = 1 \mid \tilde{X}_i = 1) = 1 - \frac{\epsilon}{3}$

A simple example (Local Differential Privacy)



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- Higher η, higher privacy, worst estimation

$$\mathrm{Var}[\hat{p}] = p(1-p) + \frac{1}{4}\left(\frac{1}{(1-\eta)^2} - 1\right)$$

More elaborated example (Differential Privacy)

- Same framework $X_i \in \{0, 1\}$, Bernoulli parameter $p \approx \frac{1}{n} \sum_{i=1}^{n} X_i$
- Data are stored (say, by hospital) but can be queried to learn p
  - An "algorithm" (say, insurance company) wants to estimate p
  - By querying estimate of p on admissible sub-groups (say, $\{i \leq K\}$)
  - Prevent it to learn precisely X_i
- Design responses to queries not "too dependent" on each X_i
  - If $n = 1000$, dependency of $\bar{p}$ on X_i is (of order) 0.1%
  - Same dependency but with n = 100 ?

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  - Same dependency but with n $= 100$?
- $\tilde{p}_D = \frac{1}{D} \sum_{i \in D} X_i + \eta$ with η noise (large variance)
  - Laplace Mechanism: η has density $\propto exp(-\epsilon \mid \eta)$

$$\left| \log \mathbb{P} \left\{ A(D) \in Z \right\} - \log \mathbb{P} \left\{ A(D') \in Z \right\} \right| \leq \epsilon$$

  - D and D' differ by only one element

Differential privacy concepts

1. ϵ-differential privacy (the second example).
   - An algo A (maps datasets to S) is ϵ-differential private if

$$\mathbb{P}\left\{A(D) \in S\right\} \leq \mathbb{P}\left\{A(D') \in S\right\} e^{\epsilon}$$

   for every $S \subset S$ and data-sets D, D' differing by one element.
2. ϵ-local differential privacy (the first example)
   - An algo A (maps user-data to S) is ϵ-locally differential private if

$$\mathbb{P}\left\{A(x) \in S\right\} \leq \mathbb{P}\left\{A(x') \in S\right\} e^{\epsilon}$$

   for every $S \subset S$ and possible pair (x, x') of user data

Micro-eco/game theoretic version of privacy

- User/Agent/Company with private type $\theta \in \Theta$

  Prior $\theta \sim \pi_0 \in \Delta(\Theta)$
  known by everyone (=environement)

  Decision space: χ
  A new (company) policy, some (user) action, etc.

  Utility $u_\theta(x) : \chi \mapsto \mathbb{R}$
- Strategy/Behavior random decision $x(\theta) : \Theta \mapsto \Delta(\chi)$
  - Environment observes public decision (not type)
  - Posterior $\pi_1 \in \Delta(\theta)$ (Bayes formula)
- Loss of privacy: "distance" between π_0 and π_1 (KL, f-divergence)

$$\max_{x(.)} \mathbb{E}_{\theta \sim \pi_0} \mathbb{E}_{\theta \sim x(\theta)} \left[u_\theta(x) \right] \text{ s.t } \mathbb{E} KL(\pi_0, \pi_1) \leq \epsilon$$

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General model and first notion

- Focus on Binary classification
  - $\hat{Y} \in \{\pm 1\}$ predicts $Y \in \{\pm 1\}$ based on feature $X \in \chi$
  - Sensible attribute $A \in \{a, b\}$
- "Fair" algorithm w.r.t. the sensible variable A
  - Many different notions of fairness
  - Incompatible and/or irreconcilable
- First, natural (?) concept

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Independence

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Independence

$$\mathbb{P}\left\{\hat{Y} = 1 \mid A = a\right\} = \mathbb{P}\left\{\hat{Y} = 1 \mid A = b\right\}$$

- What if Y is correlated to A (before or after "selection")

Two other, refined, concepts

- Independence(of $\hat{Y}$ and A) $\mathbb{P}\left\{\hat{Y} = 1 \mid A = a\right\} = \mathbb{P}\left\{\hat{Y} = 1 \mid A = b\right\}$
  - But Y correlated to A
- Independence of $\hat{Y}$ and A conditionally to Y

Separation

$$\mathbb{P}\left\{\hat{Y} = 1 \mid A = a, Y = y\right\} = \mathbb{P}\left\{\hat{Y} = 1 \mid A = b, Y = y\right\}$$

- Independence of Y and A conditionally to $\hat{Y}$

Sufficiency

$$\mathbb{P}\left\{Y = 1 \mid A = a, \hat{Y} = y\right\} = \mathbb{P}\left\{Y = 1 \mid A = b, \hat{Y} = y\right\}$$

  - If the prediction $\hat{Y}$ already subsumes A

Relations / Impossibility between concepts

1. If A & Y not independent, then independence and sufficiency cannot hold simultaneously
2. If A & Y not independent and $\hat{Y}$ & Y not independent, then independence and separation cannot hold simultaneously
3. If A & Y not independent and all values of (A,Y,$\hat{Y}$) have positive proba, then sufficiency and separation cannot hold simultaneously

Achieving Fairness

1. Pre-processing: Try to get the other features independent of A (cons: A and Y can be correlated)
2. Training: impose fairness as a constraint (cons: must have access/modify the learning algo.)
3. Post-processing Adjust a classifier to be uncorrelated with A (cons: modification of learned classif. additional risk)

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Separation can be achieved with scoring rules

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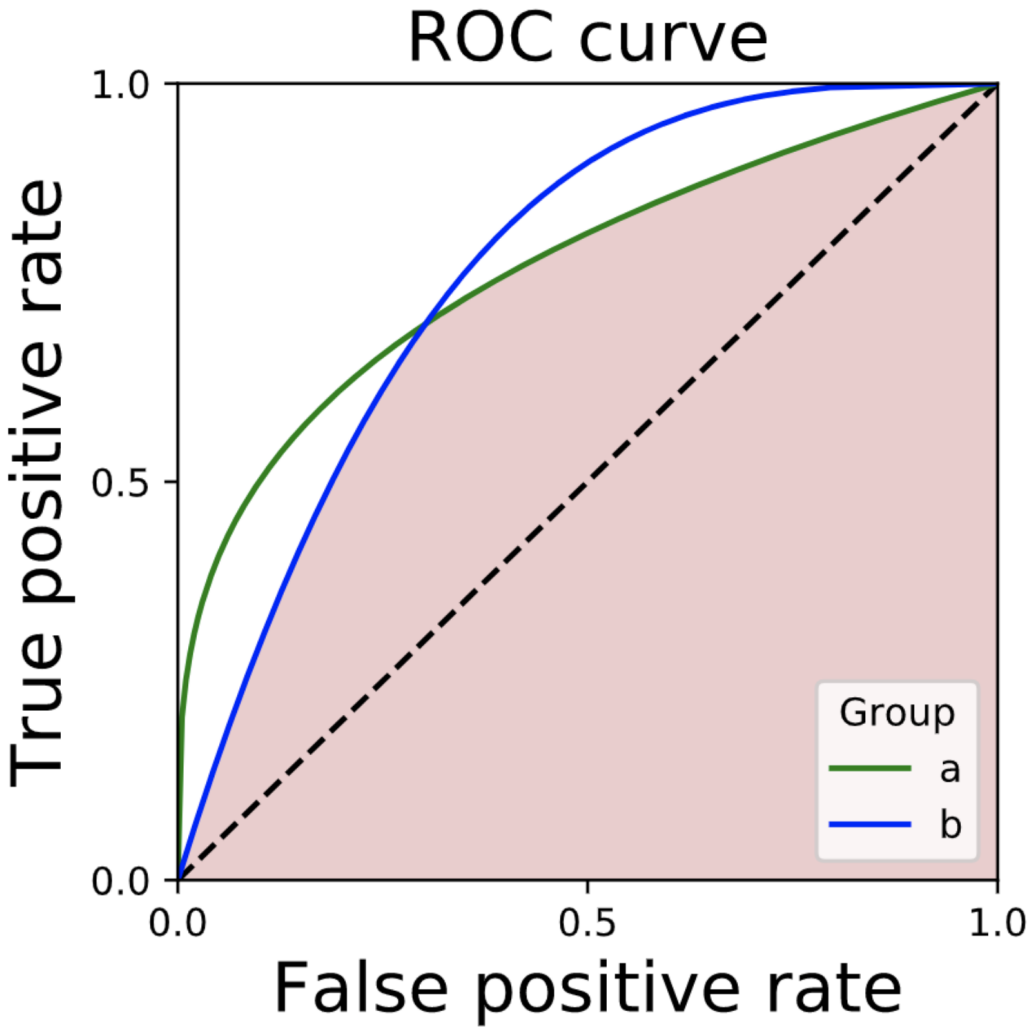

Additional concepts and limitations

1. (Conditional) Group Fairness
   - $\mathbb{P}\left\{\hat{Y} = 1 \mid A = a, S\right\} = \mathbb{P}\left\{\hat{Y} = 1 \mid A = b, S\right\}$
   - Equivalent to Independence if $S = \Omega$
2. Equal Opportunity
   - $\mathbb{P}\left\{\hat{Y} = 1 \mid Y = 1, A = a\right\} = \mathbb{P}\left\{\hat{Y} = 1 \mid Y = 1, A = b\right\}$
3. Fairness and Privacy not (always) compatible
   - One needs data, the other "hides" them

Take Home Message

- **Privacy**: data of a specific user can-not be "recovered"

ϵ-differential privacy

$$\mathbb{P}\{A(D) \in S\} \leq \mathbb{P}\{A(D') \in S\} e^{\epsilon}$$

  - Replace user data by uniform noise with small proba
  - Add (Laplace) noise to the output.
- **Fairness:** sensitive data of a user should not influence algo

Separation

$$\mathbb{P}\left\{\hat{Y} = 1 \mid A = a, Y = y\right\} = \mathbb{P}\left\{\hat{Y} = 1 \mid A = b, Y = 1\right\}$$

  - Many other definitions, all incompatible with each other
  - Separation achieved by post-processing (ROC curves)
- **Privacy** and **Fairness** not always compatible